

Section - I: Multiple Choice Questions

1. What happens when a machine learning model is too complex and performs well on the training data but poorly on new unseen data?

- A) Underfitting
- B) Overfitting
- C) Bias-Variance Tradeoff
- D) High Variance

2. What is the main difference between overfitting and underfitting in machine learning models?

- A) Overfitting occurs when a model is too complex, while underfitting occurs when a model is too simple.
- B) Overfitting occurs when a model is too simple, while underfitting occurs when a model is too complex.
- C) Overfitting occurs when a model performs well on the training set, while underfitting occurs when a model performs well on the test set.
- D) Overfitting occurs when a model performs well on the test set, while underfitting occurs when a model performs well on the training set.

3. What is the primary difference between overfitting and underfitting in machine learning models?

- A) Overfitting occurs when a model is too simple, while underfitting occurs when a model is too complex.
- B) Overfitting occurs when a model is too complex, while underfitting occurs when a model is too simple.
- C) Overfitting is a type of bias, while underfitting is a type of variance.
- D) Overfitting is a type of variance, while underfitting is a type of bias.

4. What is a common indication of overfitting in a machine learning model?

- A) The model performs well on both training and testing data
- B) The model performs well on training data but poorly on testing data
- C) The model performs poorly on both training and testing data
- D) The model requires a small amount of data to train

5. What is the main problem associated with a machine learning model that has a low training error but a high testing error?

- A) Underfitting
- B) Overfitting
- C) High bias
- D) High variance

Section - II: Descriptive / Case Study-Based Questions

1. A marketing company, "TargetTech", wants to develop a machine learning model to predict the likelihood of customers purchasing their new product based on demographic features such as age, income, and location. They collect a dataset of 1000 customers, including the target variable (purchased or not). After training a model on this dataset, they observe the following performance metrics: - Training accuracy: 98% - Testing accuracy: 60% What type of problem is TargetTech likely facing in their machine learning model, and how would you suggest they address it?
2. A marketing company, "TargetPro", uses machine learning models to predict customer churn. They have collected a dataset of 10,000 customer records, each with 20 features, such as age, location, and purchase history. The company's data scientist, Emma, trains a neural network model on 80% of the data and evaluates its performance on the remaining 20%. The model achieves an accuracy of 98% on the training data but only 60% on the test data. Emma's colleague, Ryan, suggests that the model is overfitting and recommends increasing the training dataset size to reduce overfitting. However, Emma is concerned that collecting more data may not be feasible due to resource constraints. Explain the possible reasons for the disparity in the model's performance on the training and test data. Suggest two alternative strategies that Emma could use to address the issue without increasing the dataset size.
3. A marketing company wants to develop a machine learning model to predict customer churn based on various factors such as age, location, and purchase history. They collect a dataset of 1000 customers, with 700 having churned and 300 still active. They decide to use a neural network with 10 hidden layers to train the model. After training, they achieve an accuracy of 99% on the training dataset but only 60% on the validation dataset. What type of problem is the marketing company likely facing, and what would be a suitable approach to address this issue?
4. A marketing company, Green Earth, wants to develop a machine learning model to predict the likelihood of a customer purchasing a new eco-friendly product based on demographic and behavioral data. They collect a dataset of 1000 customers, including features such as age, location, occupation, and purchase history. The model is trained on 800 customers and tested on the remaining 200. After training the model, the company notices that it performs extremely well on the training data, with an accuracy of 95%. However, when they test the model on the remaining 200 customers, the accuracy drops to 60%. The marketing team is confused and asks for your help to explain this discrepancy. What type of problem is the marketing team likely facing, and how would you suggest they address it?
5. A marketing company, "TargetAds," uses machine learning models to predict customer responses to their ads. They have collected a dataset of customer features, such as age, location, and purchase history, along with the corresponding ad responses (clicked or not clicked). The company's data scientist, Rachel, has trained a model on this dataset and obtained the following training and testing accuracy results: Training Accuracy: 98% Testing Accuracy: 60% Rachel is concerned about the performance of the model and wants to identify the problem. She asks you to analyze the situation and provide a possible explanation for the discrepancy between the training and testing accuracy results. What could be the likely reason for

CHARUSAT University

Subject: Machine learning Total Marks: 30

this discrepancy, and how would you suggest Rachel address this issue?